FV 2

(IZ VAJE 1)

2 b)

nek popeljek (pojemek)

$\sim$

$v_1(t) = v_0 - kt^2$

$k = 0{,}1\ m/s^3$

$t_u = ?$

$v(t) = \frac{ds}{dt}$

$s(t) = \int v(t)\,dt$

Še vedno velja

$s_p(t_u) = s_1(t_u)$

$s_1(t) = \int v(t)\,dt = \int (v_0 - kt^2)\,dt$

$= v_0 t - k\frac{t^3}{3}$

(od zadnjič)

$\frac{1}{2} a_p t^2 = v_0 t - k\frac{t^3}{3}$

$\Rightarrow \frac{1}{2} a_p t - v_0 + k\frac{t^2}{3} = 0$

$a = \frac{k}{3} \quad b = \frac{1}{2} a_p t \quad c = -v_0$

$ax^2 + bx + c = 0$

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$\Rightarrow t = \frac{-\frac{1}{2} a_p \pm \sqrt{\frac{1}{4} a_p^2 + \frac{4kv_0}{3}}}{\frac{2k}{3}}$

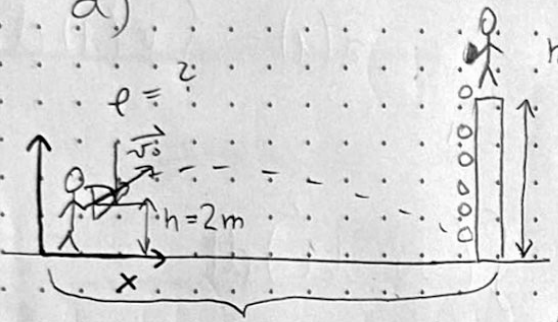
stavimo:

$$= \frac{1}{2}\,5\,m/s^2 \pm \sqrt{\frac{1}{4}\,25\,m^2/s + \frac{4}{3}\cdot 0{,}1\,m/s^3 \cdot 16{,}67\,m/s}$$

$$\frac{2}{3}\cdot 0{,}1\,m/s^2$$

DE 2:

a)



puščica zadane, ko sta ob istem času na isti točki

$l = 50\,m$ $\quad x_0 = (0, h)$

PLAN: 1. OPIŠEMO GIBANJE PUŠČICE IN GOLOBA

2. IZENAČIMO, POIŠČEMO TOČKO SREČANJA

3. IZRAČUNAMO KOT φ

$\varphi = ?$

$|\vec{v_0}| = v_0 = 100\,m/s$

PUŠČICA

$$\vec{F_p} = (0, -m \cdot g)$$

$$\vec{a_p} = \frac{\vec{F_p}}{m} = \left(\frac{0}{m}, -\frac{mg}{m}\right) = (0, -g)$$

$$\vec{v}(t) = \int \vec{a_p}(t)\,dt = \left(\int 0\,dt, \int -g\,dt\right)$$

$$= \left(0 \cdot t + v_x(t=0), -gt + v_y(t=0)\right)$$

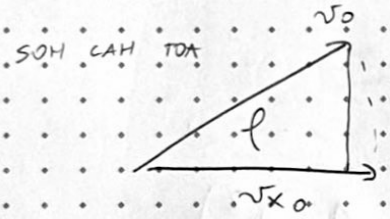
$$v(t) = at + v_0$$

$$a(t) = \frac{dv}{dt} = a + \cancel{0}$$

$$v(t) = \int a(t)\,dt = \int a\,dt = at + c$$

$$\Rightarrow = (v_{x0},\ v_{y0} - gt)$$

$$\vec{v_0} = (v_0 \cdot \cos\varphi,\ v_0 \cdot \sin\varphi)$$



$$= (v_0 \cdot \cos\varphi,\ v_0 \sin\varphi - gt)$$

$$= \underbrace{\vec{v_0} + \vec{a_p} \cdot t}_{(v(t) = v_0 + at)}$$

položaj:

$$\vec{r}_p(t) = \int \vec{v}_r(t)\,dt = \int (\vec{v_0} + \vec{a_p}t)\,dt$$

(ni odvisen od časa)

$$= v_0 \int dt + \vec{a_p} \int t\,dt + \vec{x}_0$$

$$= x_0 + v_0 \cdot t + \frac{1}{2}\vec{a_p}t^2$$

$$= (0 + v_0 \cdot \cos\varphi \cdot t + 0,\ h + v_0 \sin\varphi t - \frac{gt^2}{2})$$

② GOLOB

$\vec{r}_{x_0}^{\,g} = (l, h_0)$

$\vec{v}_0^{\,g} = (0, 0)$ ~ ker spust.

$\vec{a}_g = (0, -g)$

vrto kot preg (ker je pospešek konstanten)

$\vec{r}_g(t) = (L, h_0) + (0,0)t + (0,-g)\frac{1}{2}t^2$

$= (L, h_0 - \frac{1}{2}gt^2)$

2 ko puščica zadane goloba : t

$\vec{r}_P(t) = \vec{r}_g(t)$

$(v_0 \cos\varphi t,\ h + v_0 \sin\varphi t - \frac{1}{2}gt^2) = (L, h_s - \frac{1}{2}gt^2)$

x - smer $v_0 \cos\varphi t = L$

y - smer $h + v_0 \sin\varphi t - \cancel{\frac{1}{2}gt^2} = h_s - \cancel{\frac{1}{2}gt^2}$

$t = \frac{L}{v_0 \cos\varphi}$

$h + v_0 \sin\varphi \frac{L}{v_0 \cos\varphi} = h_s$

$\cancel{v_0} \sin\varphi \frac{L}{\cancel{v_0}\cos\varphi} = h_0 - h$

$\frac{\sin\varphi}{\cos\varphi} = \frac{h_s - h}{L}$

$\tan\varphi = \frac{h_s - h}{L}$

$$\varphi = \arctan\left(\frac{h_s - h}{L}\right) = \arctan\frac{10\,m - 2\,m}{50\,m}$$

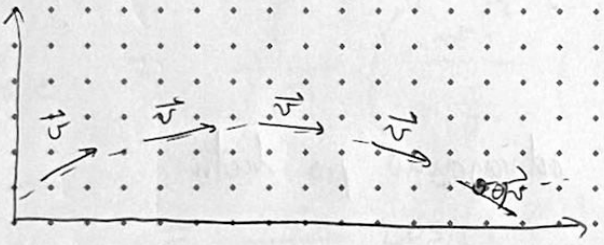
$$\doteq \underline{\underline{9{,}1^\circ}}$$

integriranje velikosti vektorjev

pilotaj

b) kot θ pod katerim puščica zadane golobu

$\theta \rightarrow$ "smer hitrosti" puščice



$$\vec{v}_p(t) \doteq (v_0 \cos\varphi\,,\ v_0 \sin\varphi - gt)$$ ← poznamo id prej

$$\doteq \left(v_0 \cos\varphi\,,\ v_0 \sin\varphi - g\frac{L}{v_0 \cdot \cos\varphi}\right)$$ (hitrost, ko puščica zadane golobu)

$v_0 \cos\varphi = v_x$

$v_0 \sin\varphi - g\frac{L}{v_0 \cdot \cos\varphi} = v_y$

θ

$|\vec{v}_p(t)|$

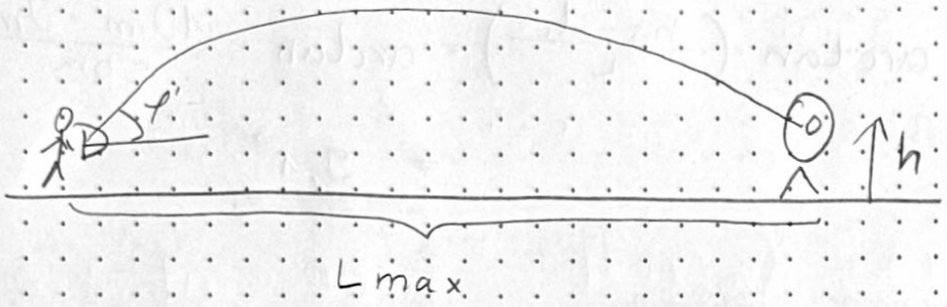
$\vec{v}_p(t)$ pri $t = \frac{L}{v_0 \cdot \cos\varphi}$

$$\Rightarrow \tan\theta \doteq \frac{v_{py}}{v_{px}} = \frac{v_0 \sin\varphi - g\frac{L}{v_0 \cdot \cos\varphi}}{v_0 \cos\varphi}$$

$$\theta \doteq \arctan \doteq \frac{100\,m/s \sin 9{,}1^\circ - 9{,}8\,m/s^2 \cdot \frac{50\,m}{100\,m/s \cdot \cos 9{,}1^\circ}}{100\,m/s \cdot \cos 9{,}1^\circ}$$

$$\doteq 6{,}3^\circ$$

c)



$\varphi' \neq \varphi$ za maksimalni doseg

$$\vec{r_p}(t) = \left(v_0 \cos\varphi' \cdot t,\ h + v_0 \sin\varphi' \cdot t - \frac{1}{2} g t^2\right)$$

$$\vec{r_T} = \left(L',\ h\right)$$

2. enačbo, 3. neč.

$$\frac{d\, r_{p.x}(t)}{d\varphi'} = 0$$

odvajamo po kotu

$$r_{p.x}(t) = L'$$

$$\vec{r_p}(t) = \vec{r_T}$$

$$x: v_0 \cdot \cos\varphi' \cdot t = L' \rightarrow$$

$$y: h + v_0 \sin\varphi' \cdot t - \frac{1}{2} g \cdot t^2 = h$$

$$v_0 \sin\varphi' \cdot t - \frac{1}{2} g t = 0$$

$$v_0 \sin\varphi' = \frac{1}{2} g t$$

$$t = \frac{2 v_0 \sin\varphi'}{g}$$

$$v_0 \cos\varphi' \cdot \frac{2 v_0 \sin\varphi'}{g} = L' = \frac{2 v_0^2}{g} \sin\varphi' \cos\varphi'$$

$$= L'(\varphi')$$

$$\frac{d r_{p,x}(t)}{d\varphi'} = \frac{dL'}{d\varphi'} = 0$$

$$\rightarrow \frac{d}{d\varphi'}\left(\frac{v_0^2}{g} 2 \sin\varphi' \cos\varphi'\right)$$

posredno odvajanje $= \frac{d}{d\varphi'}\left(\frac{v_0^2}{g} \left(\sin(2\varphi')\right)\right)$

$$= \frac{v_0^2}{g} \cos(2\varphi') \frac{d(2\varphi')}{d\varphi'}$$

$$= \frac{v_0^2}{g} \cdot \cos(2\varphi') \cdot 2 = 0$$

$$\Rightarrow \cos(2\varphi') = 0$$

$$2\varphi' = \frac{\pi}{2} \; ; \; \frac{3\pi}{2} \; ; \ldots$$

$$\varphi' = \frac{\pi}{4} = 45^\circ$$

$\sin 90^\circ = 1$

$$L_{max} = L'(\varphi') = \frac{v_0^2}{g} \sin(2\varphi') = \frac{v_0^2}{g}$$

$$= \frac{(100\,m/s)^2}{10\,m/s^2} = 1000\,m$$